VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-IITI-HT-0050

Team: Soham Prabhakar Kadu · Bhakti Ajay Kadam

Problem Context

PUBLISHED EVIDENCE · CITED

44.9 %	≈ 2 cm · 75 %	1 point	> 10×
CHANGES MADE AT STILL-OPTIMAL MOISTURE[1]	MAKERS' VISUAL 'CHANGE NOW' SIGNS[2][14]	WHERE TODAY'S MOISTURE SENSOR READS[1]	SPREAD IN REPORTED EXUDATE RATES[6]

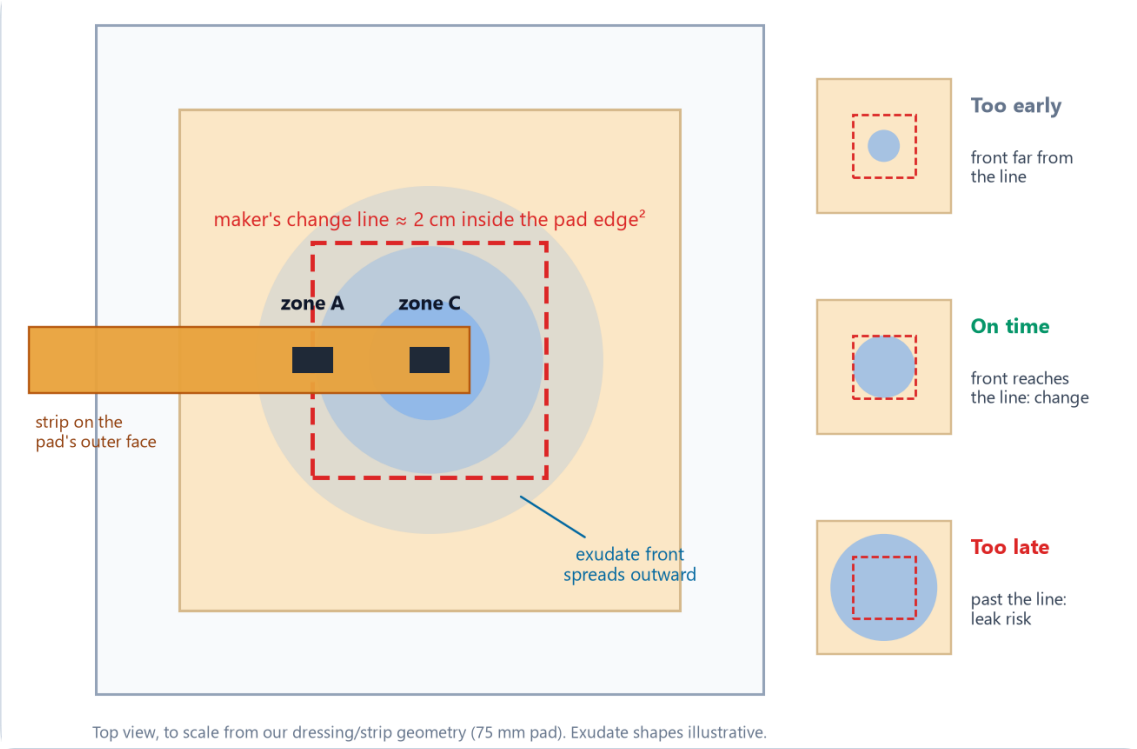


Figure: the change line is invisible under a cover (to scale, from our strip geometry).

Problem Statement & Target Beneficiary

- **Problem:** carers cannot see when a covered foam dressing reaches its maker's change line.
- **Result:** changes come too early (pain, waste) or too late (leaks, wet skin).
- **Who:** nurses and family carers of people with exuding chronic wounds: diabetic foot, leg and pressure ulcers.
- **We will:** signal that line from outside, without lifting the dressing.

Project Overview: Dermasense · Team ID VK26-IITI-HT-0050 · Health Tech: Emergency Response & Remote Monitoring · Core innovation: a strip that reads a dressing's own change line.

Problem Context: What Exists Today and the Gap

PUBLIC SEARCH · NOT EXHAUSTIVE

What a carer can use today, and what each option misses

Today's option	Without lifting?	Maker's rule?	What limits it
Fixed schedule (clock)	✗	✗	a guess; exudate varies > 10×[6]
Lift and look	✗	✓	disturbs the wound; removal hurts[20]
Visual change indicator[14]	✗	✓	works only if the dressing is visible
Single-point moisture sensor[1]	◐	✗	spot check at one point
Closed-loop smart bandage[5]	✓	✗	lab research (mice); therapy device
Diaper wetness strip[4]	✓	✗	made for diapers, not dressings
Dermasense (our target)	✓	✓	**unproven: E2 and E4 test it**

✓ **yes** ✗ **no** ◐ **partly.** Our own non-exhaustive search of papers, patents and products.

Public Search Disclosure & Our Focus

- **Searched:** papers, patents and products[14 5 14].
- **Found:** no low-cost sensor that reads a dressing's own change line (in our search).
- **Our focus:** tell the carer when exudate reaches that line, without lifting the dressing.
- **Why now:** carbon can be laser-written on polyimide in one pass[7]; makers already define 'full' by position[2 14].
- **Not our goal:** diagnosis or therapy; we only report how full the dressing is.

Design Question: how might a carer know that a covered dressing is full, without lifting it?

User Journey: Before State

CURRENT STATE · CITED

1.9–4.5 per 1,000	101 million	19–34 %	40.3 %
HAVE A CHRONIC WOUND (INDIAN SURVEYS)[16][17]	INDIANS LIVING WITH DIABETES[18]	LIFETIME FOOT-ULCER RISK IN DIABETES (EST.)[19]	PATIENTS: CHANGE PAIN IS THE WORST PART[20]

Persona (illustrative, from the cited data; not a real patient): a 60-year-old with a diabetic foot ulcer, cared for at home by family; a nurse visits to change the dressing.

STAGE	1 Dressing on	2 Days pass	3 Decide	4 Change	5 Repeat
WHAT HAPPENS	foam dressing under a bandage or sock	exudate spreads, unseen	clock, a peek, or wait for a leak	too early, or too late	about 5 times a week[13]
PAIN POINT	**the front is hidden**	**no signal**	**a guess**	**pain + waste, or wet skin**	**≈ 18 min each time[13]**

Figure: today's journey (design-thinking map; persona illustrative, numbers cited).

Quantified Inefficiency (published data)

- **Time:** ≈ 18 min per dressing visit, about 5 visits a week[13] (UK, patient-reported).
- **Waste:** 44.9 % of changes came while moisture was still optimal[1].
- **Cost:** ₹240–510 for one 10 × 10 cm foam dressing[15]; each early change discards one.
- **Safety:** a late change can leak (wet skin); removal hurts[20].
- **Who carries it:** the patient, the family carer and the nurse.

Insight: the dressing already 'knows' when it is full (its change line); the carer just cannot see it.

User Journey: After State (Intended)

INTENDED · NOT CLINICALLY VALIDATED

0 lifts	up to 44.9 %	≤ ₹100	1 in 2–5
NEEDED TO CHECK THE DRESSING (BY DESIGN)	EARLY CHANGES A RELIABLE ALERT COULD AVOID[1]	TARGET STRIP COST (ESTIMATE)	AVOIDED CHANGES PAY FOR A STRIP (PROJ.)[15]

Same persona with Dermasense (intended workflow; bench-testable, not clinically validated).

STAGE	1 Strip on	2 Reader on	3 Glance	4 Line wet	5 Remote
WHAT HAPPENS	placed on the pad's outer face	clips on: 4 pins, 2 dowels	green OK, without lifting	amber CHANGE; red CHECK if a leak	state on a nearby phone (Bluetooth, to be tested)
GAIN	**nothing on the wound**	**same routine**	**no guess, no pain**	**change on time**	**nurse sees it; reason logged**

Figure: the intended journey (design-thinking map); tested on the bench first, not on patients.

How Each Person Is Helped (intended)

- **Patient:** fewer needless removals, the step patients find most painful[20].
- **Family carer:** a clear OK / CHANGE instead of a guess.
- **Nurse:** checks without unwrapping; every change has a logged reason.
- **Budget:** fewer wasted dressings; strip ≤ ₹100, reader reused (projection).

Design Principle: same routine, nothing touches the wound, only film and carbon are disposable. Target: alert within ±10 % of the visible front (E4).

System Architecture & Specs

DESIGNED · LAB TEST PENDING

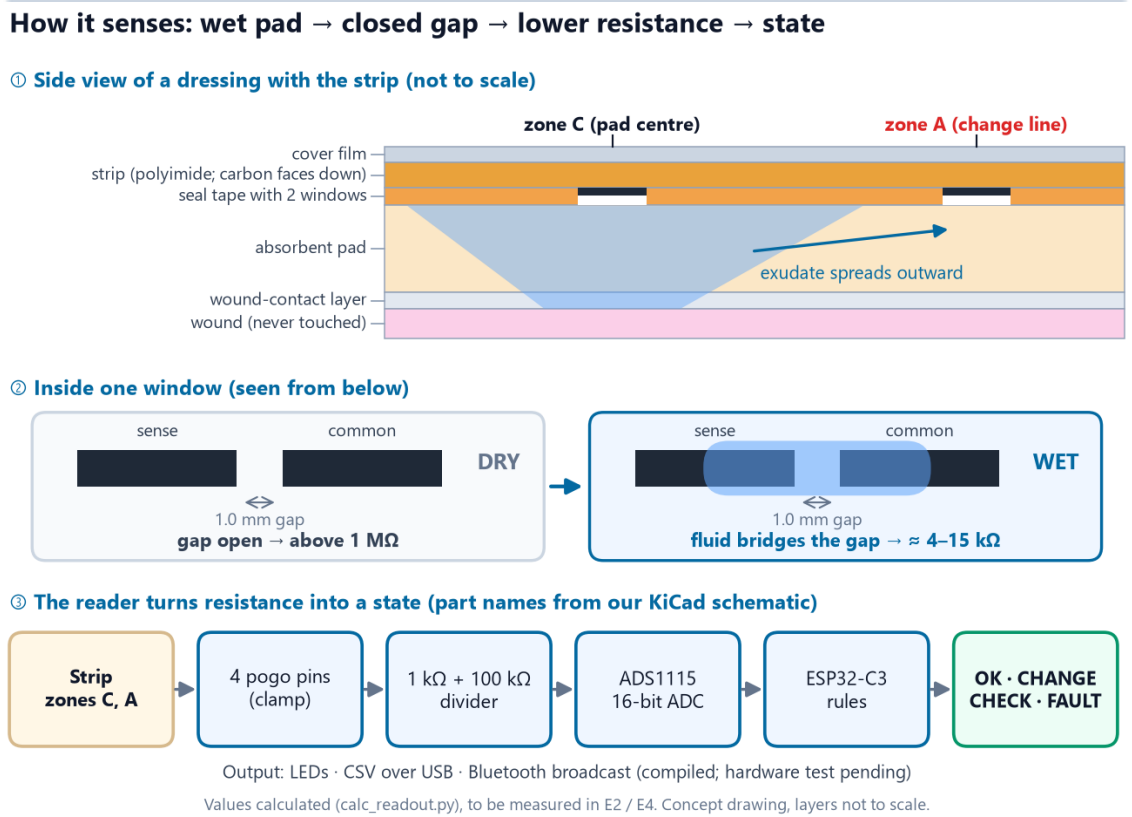


Figure: how a wet pad becomes a state (concept drawing; values calculated, to be measured).

Target Specifications & Hardware

Parameter	Design value	Basis
Sensing zones	2: centre C, change line A	decision D-012
Electrodes	1.0 mm lines and gaps	E2 confirms
Wet / dry	< 100 kΩ / > 300 kΩ, 3 scans	calc: wet 4–15 kΩ
Drive	bipolar, ≤ 33 µA, no net DC	calc · bench T8
ADC / MCU	ADS1115 16-bit (I²C) · ESP32-C3	datasheets
Self-test	100 kΩ check ±5 % + strip loop	else FAULT
Contacts	4 pogo pins + 2 dowels	CAD checked
Output	LEDs · USB log · BLE broadcast	BLE compiled only

Architecture Rationale: only film and carbon are disposable; every electronic part is reused.

System Architecture & Specs: Firmware & Data Flow

VERIFIED IN SOFTWARE · HARDWARE PENDING

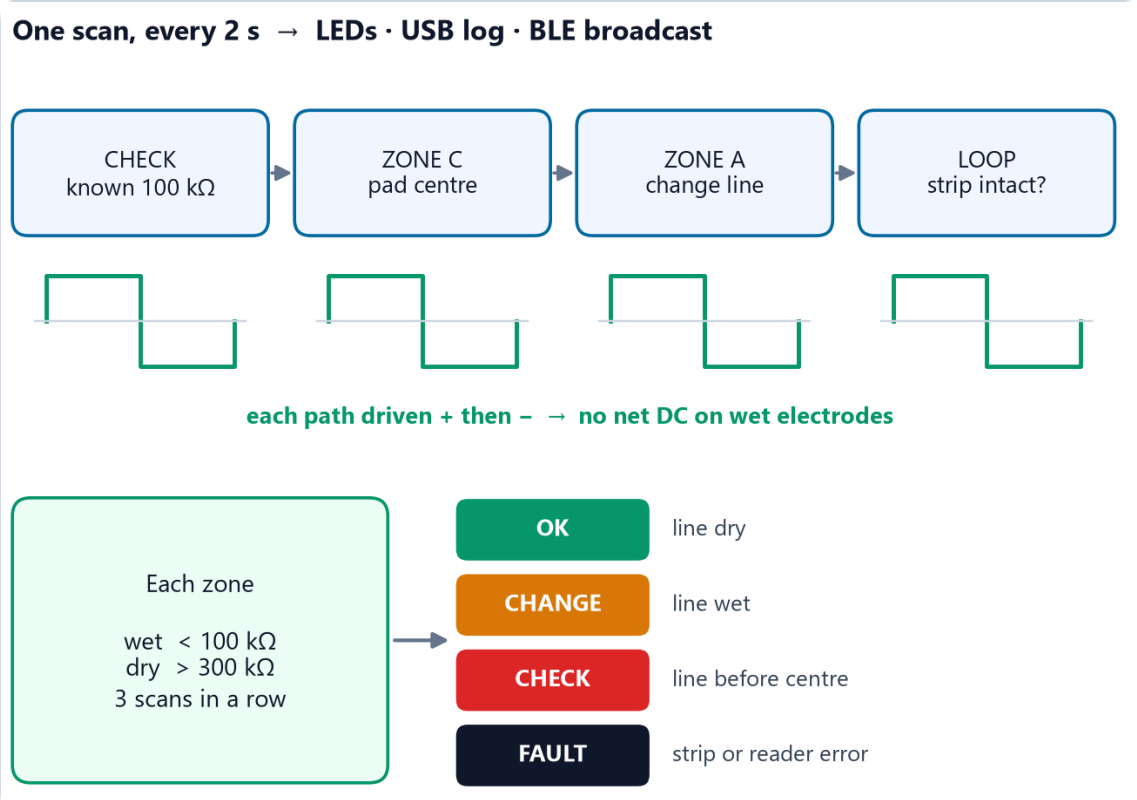


Figure: one reader scan, from measurement to state.

Firmware Logic & Data Flow

- **Measure:** each zone driven both ways, so no net DC.
- **Read:** 16-bit ADC, scaled to the 3V3 rail.
- **Decide:** wet < 100 kΩ, dry > 300 kΩ, 3 scans in a row.
- **Self-check:** a known 100 kΩ every scan, else FAULT.
- **Output:** LEDs; a CSV line over USB; a Bluetooth broadcast any scanner app can read (no custom app yet).
- **Why remote:** a carer nearby can see the state without unwrapping or plugging in (to be bench-tested).

Software vs Hardware: 23 compile-time and 16 PC tests pass; not yet run on a wired board.

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT

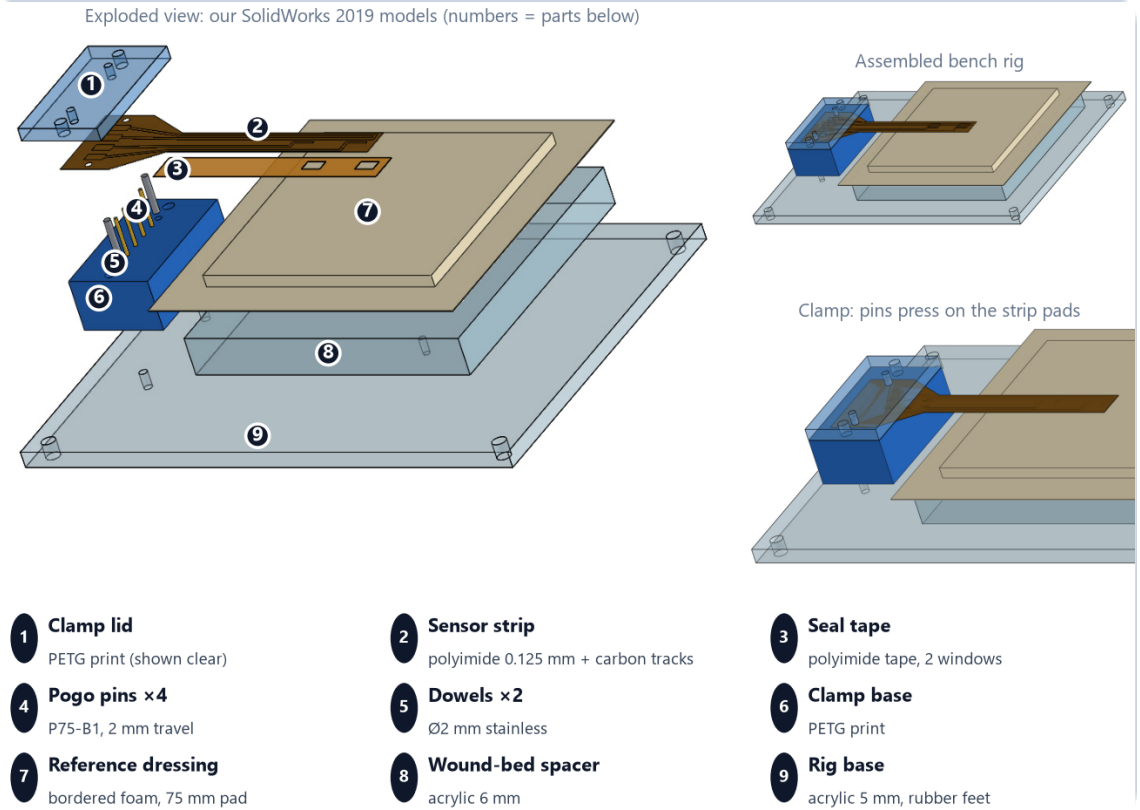


Figure: our SolidWorks 2019 models (9 parts, 1 assembly, 7 A3 drawings, STEP); not yet built.

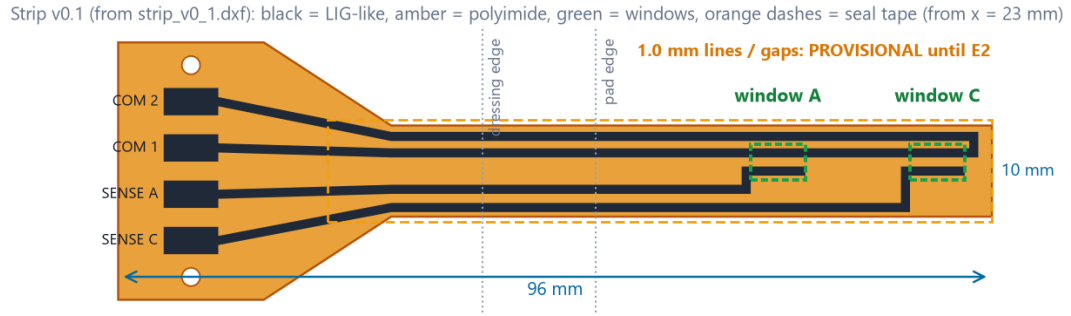
A clip-on clamp: 4 spring pins, 2 dowels

Check	Result	Basis
Materials	polyimide strip · PETG clamp · acrylic rig	printable, laser-cut
Pin force	0.72 N each · 2.9 N total	calc
Lid bending	11 µm vs 1.48 mm pin travel	calc
Alignment	0.49 mm used of ±1.35 mm	calc · 2.8× margin
Strip bending	0.62 % strain at 10 mm radius	calc · yield ≈ 3 %
CAD checks	99 pass · 0 fail · 5 wait for E2	verify_cad.py
FEA	not run: loads are under 3 N	hand calcs above

Design Iteration: a clash check found the seal tape trapped under the clamp (9.0 mm³); moved to x = 23 mm, now 0 clash (decision D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · LAB TEST PENDING



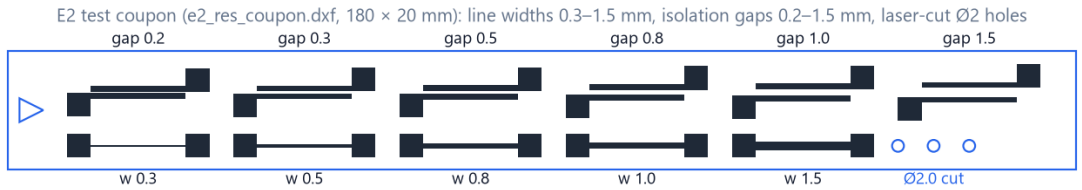


Figure: our laser files, drawn from the DXFs. Not yet lased.

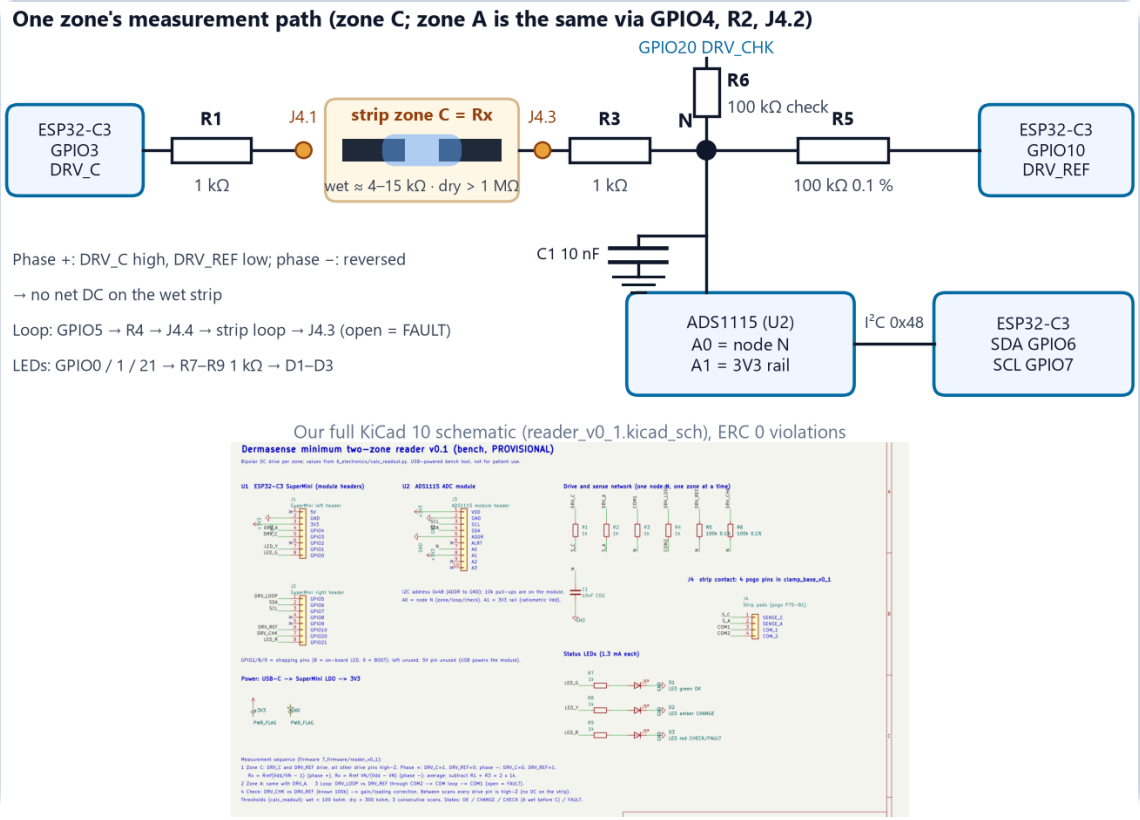
Fabrication & Process Validation (E2)

- **Material:** laser-carbonised, LIG-like layer on polyimide[7].
- **Reference:** setting '12', most likely % PWM, not watts.
- **E2 plan:** 7 coupons, 2 days, 72 pre-registered samples.
- **Pass rules:** CV ≤ 20 %, gaps > 3 MΩ, fixed beforehand.
- **Sealed:** leads under tape; LIG made in air wets easily[9].

Laser Files Checked: 35/35 file checks pass (files only; nothing lased yet). LaserCAD import is checked at the machine.

Electronics Schematics & BOM

ERC VERIFIED · NOT YET WIRED



Our full KiCad 10 schematic (reader_v0_1.kicad_sch), ERC 0 violations

Figure: zone-C path drawn from our netlist, above our real KiCad 10 sheet.

Pin & Net Map (from the KiCad netlist)

Net	Pin	Function
DRV_C · DRV_A	GPIO3 · GPIO4	drive zone C · A via 1 kΩ
DRV_LOOP	GPIO5	strip loop (broken strip)
DRV_REF	GPIO10	100 kΩ 0.1 % reference
DRV_CHK	GPIO20	100 kΩ self-check
SDA · SCL	GPIO6 · GPIO7	I²C to ADS1115 (0x48)
N · 3V3	ADS1115 A0 · A1	node · rail (ratiometric)
LED G · Y · R	GPIO0 · 1 · 21	OK · CHANGE · CHECK/FAULT
Strip pads	J4.1 – J4.4	S_C · S_A · COM1 · COM2
Power	USB-C → 3V3	LDO on the SuperMini

Verification: ERC 0 violations; netlist matches firmware and strip pads (0 mismatches).

Electronics Schematics & BOM: Bill of Materials

3 PRICES QUOTED · NOTHING ORDERED

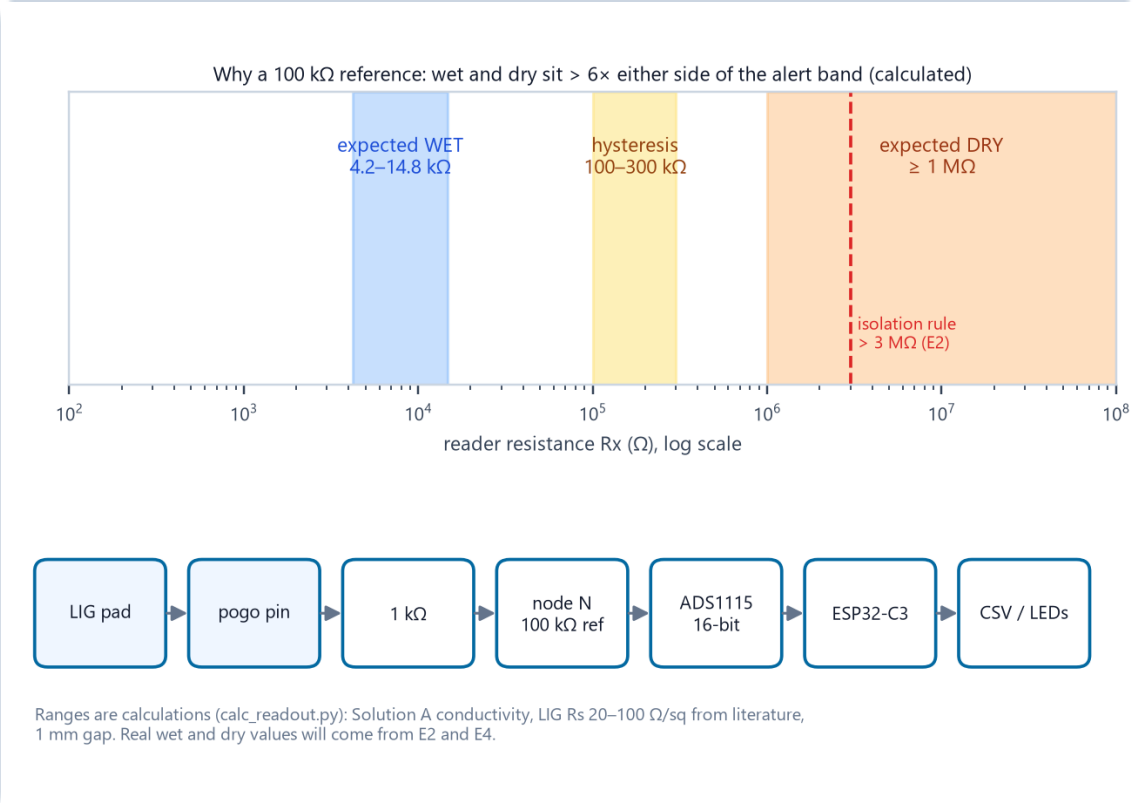


Figure: why a 100 kΩ reference (calculated ranges).

Prototype Bill of Materials

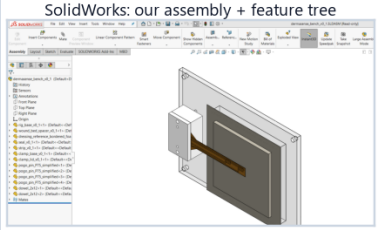
Item	Qty	Source	₹
ESP32-C3 SuperMini (PRO4607)	1	Probots · quoted	242
ADS1115 module (PRO789)	1	Probots · quoted	125
P75-B1 pogo pins (05880)	100	Hubtronics · quoted	354
Resistors, C0G cap, LEDs, board	set	local / online	158–421
Clamp + bench rig	set	lab / local	100–1,240
Strip materials	set	lab / online	800–3,500
Test consumables	set	local	480–1,060
Total			**≈ 2,259–6,942**

Procurement Status: reader parts and pins quoted 19 Oct, in stock in India; the rest are estimates. The reader is reused; only the strip is consumed.

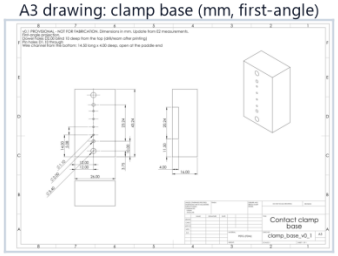
Maker Journey

REAL FILES · RE-CHECKABLE · NO STAGED PHOTOS

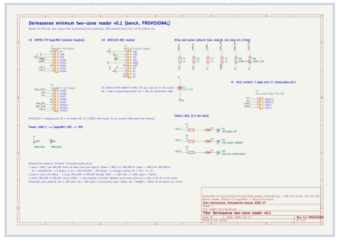
SolidWorks: our assembly + feature tree



A3 drawing: clamp base (mm, first-angle)



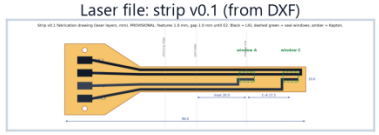
KiCad 10: reader schematic



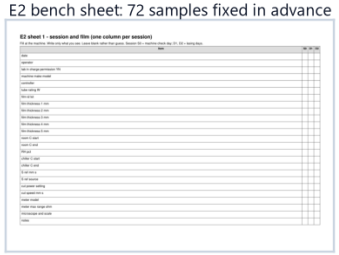
Firmware: state rule + compile-time test

```
// C = pad centre, A = maker's change line
constexpr State classify(Wet c, Wet a,
                         bool fault) {
  return fault ? State::Fault
    : (a == Wet::WET ? (c == Wet::WET
         ? State::Change : State::Check)
       : State::Ok);
}
static_assert(classify(Wet::WET, Wet::WET,
   false) == State::Change,
   "front reached the change line");
```

Laser file: strip v0.1 (from DXF)



E2 bench sheet: 72 samples fixed in advance



Checks re-run on 19 Oct 2026 (results as printed by each tool)

verify_cad.py (SolidWorks)	{'PASS': 99, 'FAIL': 0, 'BLOCKED': 5}
kicad-cli sch erc	Found 0 violations
verify_laser_files.py	35 PASS, 0 FAIL
arduino-cli compile (esp32c3)	Sketch uses 650349 bytes (49%)
pytest (firmware tools)	16 passed
pytest (E2 analysis, synthetic data)	3 passed

Figure: our own files and tool output (checks re-run 19 Oct 2026).

Our Team at Work

TEAM PHOTO

both members, institutional ID badges visible
replace: right-click → Change Picture

LAB PHOTO

the team at work (CAD, wiring or the laser)
replace: right-click → Change Picture

Soham Prabhakar Kadu · role: **[team to add]**
Bhakti Ajay Kadam · role: **[team to add]**

- **Real files:** .SLDPRT .SLDASM .SLDDRW .STEP .kicad_sch .ino .dxf, dated in git since 14 Oct.
- **Re-checkable:** one command re-runs each check shown on the left.
- **How we work:** scripts written with AI assistance (Claude), directed by us, build the CAD, schematic and firmware; the physical build and tests are done by us, starting with E2.
- **Photos:** never staged or generated.

Raw Repository Linkage: [team to insert the Drive link before upload]: all native files and a dated work history.

Maker Journey: Engineering Iterations

DOCUMENTED DECISIONS

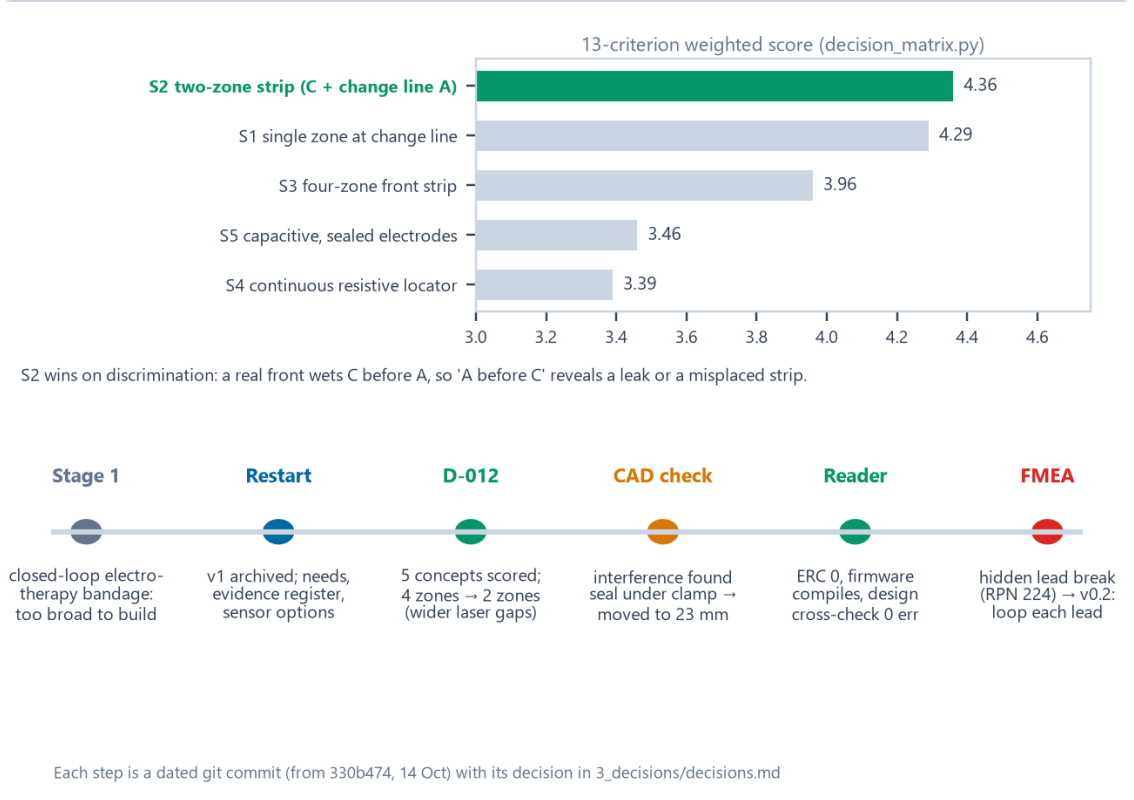


Figure: concept scores and the iteration path (each a dated commit).

Stage 1 to Stage 2 Iterations

- **Pivot:** therapy bandage (already shown in labs[5]) → one job: is it full?
- **Choice:** two-zone strip (4.36) beat one zone (4.29): it spots leaks.
- **Kept:** Stage 1's Bluetooth telemetry, now a state broadcast (compiled; hardware test pending).
- **Corrected:** Stage 1 called Kapton 'medical-grade'; DuPont cautions otherwise[11].
- **Redesign:** FMEA risk → v0.2 loops each sense lead.

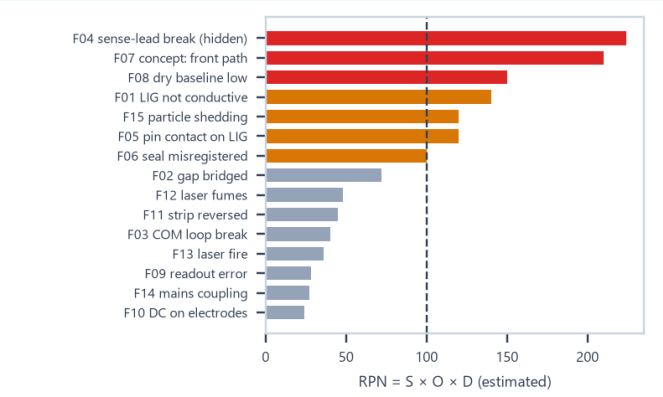
Prior Art: diaper strips[4] and visual change indicators[14] exist; our novelty is reading the change line electrically, with an order check.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

Design FMEA: top 5 of 15 failure modes (ratings estimated before tests)

Failure mode	Effect	RPN	Fix / test
F04 cracked sense lead	zone reads 'dry' for ever	224	v0.2 loops each lead · T-MECH-01
F07 front path differs	CHANGE late or never	210	E4 on real foam dressings
F08 humid dry baseline	false 'wet'	150	measure baselines (E4)
F01 weak carbon layer	zones always 'dry'	140	E2 + per-strip QC record
F05 pin contact on carbon	readings drift high	120	rounded pins · T-CON-01



Fail-safes

- **Bench only:** USB reader, never on a person.
- **No net DC:** bipolar drive, idle pins off; bench test T8 checks < 5 mV.
- **Self-check:** every scan, else FAULT.
- **No wound contact:** strip on the pad's outer face, leads sealed.
- **Laser work:** signed safety checklist every session.

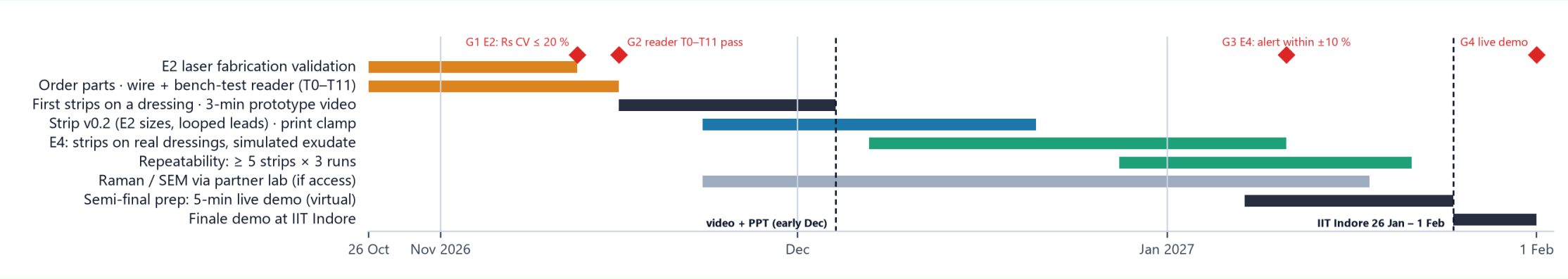
Standards (considered, not claimed)

- **ISO 14971** risk management
- **IEC 60601-1** future on-body reader
- **ISO 10993-1** before any skin contact[11]
- **EN 13726** simulated exudate for tests[10]
- **IEC 60825-1** laser work during fabrication

Roadmap

FUTURE SCOPE · GATED PLAN

4-Month Plan on the programme's own dates: each gate must pass first



Build & Bench (Months 1 - 2)

- **Nov:** E2 laser tests (G1); build and bench-test the reader (G2).
- **Early Dec:** first strips on a dressing; 3-min prototype video.

Validate & Demo (Months 3 - 4)

- **Dec:** strip v0.2 + printed clamp; E4 wet tests (G3).
- **Jan:** repeatability; virtual semi-final with a live demo.
- **Stage 3:** working strip + reader: OK → CHANGE live on a dressing, raw data logged.

Technical Challenges & Risks

FOUND IN OUR OWN VERIFICATION

Geometry: Seal Under Clamp

Found: seal tape overlapped the clamp (9.0 mm³).

Impact: would lift the strip off the pins.

How: automated clash check.

Circuit: Hidden Lead Break

Found: a cracked sense lead reads 'dry'.

Impact: CHANGE never shown, no alarm.

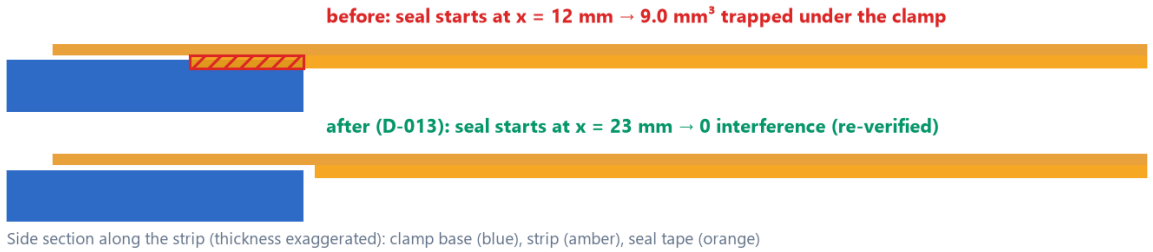
How: design FMEA (RPN 224).

Process: Unknown Laser Recipe

Found: '12' is likely a % command; tubes are unsteady at low power[8].

Impact: every strip size is provisional.

How: LaserCAD config audit.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



Also Caught: STEP exported in cm (now mm); a drawing showed 0.13 for 0.125 mm (fixed).

Potential Solutions & Risk Mitigation

FIXED · DESIGNED · FUTURE

Fixed & Verified

Seal: starts at x = 23 mm; 0 clash.

Units: all STEP files in mm.

Checks: 99 CAD checks pass.

Designed, to Prove

v0.2 strip: every sense lead looped.

Contact: silver pads, rounded pins.

Self-test: check resistor → FAULT.

Fallbacks

Wider lines: if E2 says so.

Process: slower passes, log current.

Concept: one-zone strip (score 4.29).

Future Scope & Validation Plan

- **Experiments:** E2 laser, reader bench test, E4 wet tests.
- **Material:** Raman and SEM before any 'graphene' claim.
- **Reliability:** bending, contact cycling, humidity drift.
- **Product:** battery pod; phone app on the BLE broadcast; strips per dressing.
- **Before patients:** ISO 10993, IEC 60601-1, ethics approval.
- **Not claimed:** sensing performance, graphene, clinical benefit.

Rule We Keep: no performance, graphene or clinical claim until it is measured.

References & Sources

SOURCES WITH READING DEPTH

Numbers match the superscripts on each slide. After each source we state how deeply it was read; nothing is labelled verified beyond that.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read

2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts

3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract

4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read

5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage

6. Dealey C., Cameron J., Arrowsmith M., A study comparing two objective methods of quantifying the production of wound exudate, J. Wound Care 15(4):149–153 (2006) · abstract (methods disagree)

7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read

8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed

9. Li Y. et al., Laser-induced graphene in controlled atmospheres: from superhydrophilic to superhydrophobic surfaces, Adv. Mater. 29:1700496 (2017). doi:10.1002/adma.201700496 · abstract

10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network

11. DuPont position statement on the use of DuPont materials in medical applications (Kapton caution) · web page

12. Texas Instruments ADS1115 datasheet (SBAS444); Espressif ESP32-C3 datasheet · vendor documents

13. Wounds UK news (22 Aug 2018) on a Mölnlycke-commissioned survey of >200 UK patients with chronic wounds: average dressing time 18 min · patient-reported, industry-sponsored; web page read

14. Smith+Nephew, ALLEVYN LIFE product page: visual change indicator ('At 75%, it's time to change your dressing') · manufacturer web page, read 19 Oct 2026

15. Truemeds.in retail listings: Mepilex Lite 10 × 10 cm ₹242; Mepilex Border 10 × 10 cm ₹514 (pack size unclear) · retail pages, read 19 Oct 2026

16. Gupta N., Gupta S.K., Shukla V.K., Singh S.P., An Indian community-based epidemiological study of wounds, J. Wound Care 13(8):323–325 (2004). doi:10.12968/jowc.2004.13.8.26657 · abstract

17. Sharma A. et al., Epidemiological profile of chronic wounds in an Indian population, Cureus 16(9):e68684 (2024). doi:10.7759/cureus.68684 · full text

18. Anjana R.M. et al., ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), Lancet Diabetes Endocrinol. 11(7):474–489 (2023) · abstract and summaries

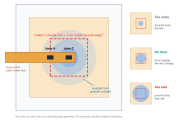
19. Armstrong D.G., Boulton A.J.M., Bus S.A., Diabetic foot ulcers and their recurrence, N. Engl. J. Med. 376:2367–2375 (2017). doi:10.1056/NEJMra1615439 · summaries; range is an estimate and contested

20. Price P.E. et al., Dressing-related pain in patients with chronic wounds: an international patient perspective, Int. Wound J. 5(2):159–171 (2008). doi:10.1111/j.1742-481X.2008.00471.x · abstract

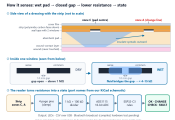
Image Sources

ALL IMAGES FROM OUR FILES

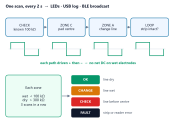
Every image comes from our own project files: SolidWorks renders and screenshots, the KiCad export, and plots drawn by our scripts (written with AI assistance). No stock photos and no AI image generators. Team photos (slide 12) are taken by the team.



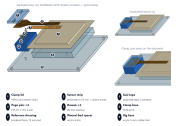
Problem illustration, drawn to scale
Source: make_figures.py · strip_v0_1_geometry.json
1_presentation/figures/fig_problem.png



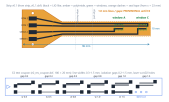
How it senses (concept)
Source: make_figures.py · sensor_selection.md, calc_readout.py
1_presentation/figures/fig_howitworks.png



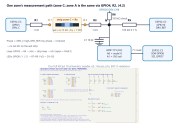
Firmware scan and state logic
Source: make_figures.py · reader_v0_1 source
1_presentation/figures/fig_dataflow.png



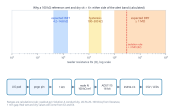
Annotated SolidWorks views
Source: render_cad.py + make_figures.py · 5_cad/solidworks
1_presentation/figures/fig_cad.png



Strip and E2 coupon layers
Source: make_figures.py · our DXF files
1_presentation/figures/fig_strip.png



Measurement path + KiCad sheet
Source: make_figures.py · reader_v0_1 netlist + KiCad PDF
1_presentation/figures/fig_schematic.png



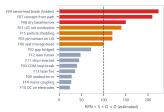
Measurement window
Source: make_figures.py · calc_readout.py
1_presentation/figures/fig_window.png



Evidence gallery + tool output
Source: our files, screen capture, 19 Oct tool runs
1_presentation/figures/fig_maker.png



Decision matrix + iterations
Source: make_figures.py · decision_matrix.md
1_presentation/figures/fig_decision.png



FMEA risk ranking
Source: make_figures.py · fmea_v0_1.csv
1_presentation/figures/fig_fmea.png



4-month roadmap
Source: make_figures.py · E2/bench/E4 plans
1_presentation/figures/fig_roadmap.png



Seal clash + lead-break diagrams
Source: make_figures.py · geometry + FMEA
1_presentation/figures/fig_challenges.png